

Md. Kaif Faiyaz

[kaif-github](#)

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| [Kaif Faiyaz](#)

Kolkata, West Bengal, India

Education

• Indian Institute of Technology Kanpur

2019 - 2024 Kanpur, India

BS-MS Dual Degree in Physics; Minor in Industrial Management & Engineering

– CGPA: 8.42/10 (BS), 8.9/10 (MS)

Research Paper

• Multihead Adaptive Genetic Algorithm in Non-Stationary Environments

May'24-Nov'25 (SUBMITTED)

with Prof. Tushar Sandhan (under review at *IEEE Transactions on Cybernetics*)

[Paper](#)

- Recast evolutionary search as a controlled stochastic dynamical system for non-stationary financial environments.
- Derived warmstart survivorship bounds using Bernstein-Serfling concentration and threshold-exceedance arguments.
- Analyzed selection dynamics via the Price equation and ECDF-based approximations for intractable fitness probabilities. Validated the framework through ablations on indices and FX, showing robust walk-forward performance improvements.

• Electrically driven quantum metric Bloch oscillations

April'23 - April'24 (SUBMITTED)

with Prof. Amit Kumar Agarwal (under review at *Physics Review Letter-B*)

[Paper](#)

- Developed a modified semiclassical framework to analyze electron wave-packet dynamics under inhomogeneous electric fields, deriving closed-form expressions for real-space oscillations driven by the quantum metric.
- Conducted analytical and asymptotic derivations of drift and oscillatory transport responses-separating intrinsic vs. extrinsic mechanisms-and established scaling laws for current saturation and differential conductance.
- Implemented simulations on tilted Dirac models, performing parameter sweeps over field strength, chemical potential, and tilt to validate theoretical predictions and quantify quantum-metric-induced transport behaviors.

Self-directed Research

• Finite-Sample Theory for Causal Tail Dependence Estimators

Nov'25 - Present

Manuscript in preparation

[Project](#)

- Derived explicit finite-sample error bounds for nonparametric estimators of the upper tail dependence coefficient, combining Bernstein concentration inequalities with second-order tail bias control.
- Analyzed rank-based estimators under unknown marginals using DKW-type uniform bounds, quantifying the additional error induced by marginal estimation. Established bias-variance tradeoffs and optimal threshold selection for tail dependence estimation, with validation via copula-based simulations and real hydrological data.

Research Projects

• Modeling of Tail Dependence in Airline Stock Prices

Aug 2023 - Jan 2024

Supervisor: Prof. Arnab Hazra

[Project](#)

- Modeled joint upper- and lower-tail dependence in Indigo and SpiceJet daily log-returns during COVID-19 using a flexible three-parameter copula with asymmetric Laplace components, combined with GEV-based marginal transformations and local likelihood estimation for time-varying dependence.
- Performed preprocessing via GEV-based uniformization of return-rate series with spline-selected kernel parameters using bootstrap validation, enabling stable inference of dynamic tail-dependence coefficients and asymptotic (in)dependence classes.
- Developed a hierarchical Bayesian model for time-varying $(\delta_l, \delta_u, \rho)$ using Kronecker-structured multivariate normal priors; derived Gibbs updates for Θ and β , implemented Metropolis-Hastings samplers for ϕ and Σ , and analyzed numerical stability issues arising from near-singular correlation matrices.

• Multi-Company-Multi-Period Demand Response Management in Smart Grid

Sep 2022 - Nov 2022

Supervisor: Prof V. Ramani(IME636A)[reproduced original paper]

[Project](#)

- Formulated a multi-period, multi-company Stackelberg game for demand-response management, deriving closed-form optimal consumer demand, company pricing, and Nash-equilibrium power allocation under constraints.
- Implemented a distributed price-update algorithm enabling to compute equilibrium strategies.
- Reproduced case studies using EcoGrid EU and Danish Smart Grid datasets, showing reduction in consumer payments by 11-36% , while also validating theoretical predictions on power allocation and equilibrium pricing.

• Application of Weibull Distribution & Weibull Process

Sep 2022 - Nov 2022

Supervisor: Prof Arnab Hazra(MTH511A)

[Project](#)

- Implemented a convex mixture of Weibull densities to model multi-seasonal wind-speed distributions, formulating and solving a highly nonlinear constrained MLE problem using Sequential Quadratic Programming (SQP).

- Performed goodness-of-fit evaluation through empirical CDF construction and two-sample Kolmogorov-Smirnov tests, demonstrating improved fit of the optimized mixture model compared to standard Weibull assumptions.
- Developed and coded a Monte Carlo EM algorithm for parameter estimation of a Weibull process under left-censored failure-time data, including simulation of missing data via inverse-transform sampling and iterative maximization of the expected log-likelihood.

• An Efficient Algorithm for Bayesian Nearest Neighbours

Jan 2022 - April 2022

Supervisor: Prof Arnab Hazra(MTH535A)[reproduced original paper]

Project

- Developed an efficient Bayesian change-point-based algorithm to compute the posterior distribution of the optimal k for k-Nearest Neighbours, avoiding MCMC and achieving order-of-magnitude faster inference than simulation-based methods.
- Implemented full probabilistic calibration of neighbourhood size by recursively evaluating change-point probabilities on distance-ordered data, and applied the method to the Ripley dataset for binary classification.
- Conducted a performance benchmarking study comparing Bayesian k selection with naïve k-NN across 1-125 neighbour regimes, demonstrating a reduction in misclassification rate from 0.104 to as low as 0.036.

Coursework

† AWARDED **A/A*** grade

Statistical Inference & Data Analysis†, Data Structures & Algorithms, Game Theory†, Stochastic Processes, Bayesian Analysis†, Probability & Statistics, Quantum Mechanics†, Statistical Mechanics, Linear Algebra, Differential Equations, Complex Analysis, Partial Differential Equations, Statistical Physics, Real Analysis & Multivariable Calculus†

Awards

- **JEE Advanced 2019** Secured **All India Rank 1419** among the 250,000+ screened candidates.
- **JEE Mains 2019** Secured **All India Rank 2286** among the 1.6 million candidates.
- **WBJEE 2019** Secured **All India Rank 63** among the 130,000+ candidates.

Work Experience

• Aviso AI

Jul 2024 - Present

Conversational Intelligence, Data Scientist-II

Remote

- Developed a privacy-preserving multimodal deception detection system using self-supervised cross-modal and temporal contrastive learning; framed as weakly supervised inference with multiple-instance learning.
- Designed a decision-support system for product recommendation and pricing using RAG models; incorporated user feedback via preference-based optimization (DPO), enabling iterative updating of model behavior.
- Proposed an importance-sampling-based optimization framework for prompt selection using LLM-as-a-judge; employed active sampling strategies to reduce variance and improve sample efficiency in model evaluation.
- Designed a bimodal speech emotion recognition framework combining embeddings with GRU-based temporal modeling; addressed class imbalance using data augmentation and loss reweighting, improving Macro-F1 by 8.03%.

• Samsung Research Institute Bangalore

May 2023 - Jul 2023

Developer Intern

Bangalore, India

- Automated **ML pipeline deployments** by orchestrating VMs with Ansible, configuring Kubernetes clusters, and managing Docker environments on remote Linux servers, streamlining scalable experimentation.

Technical Skills

- **Programming Languages & DevOps**- C++, Python, R, SQL, Docker, Kubernetes, Ansible, Linux
- **Machine Learning & Deep Learning Frameworks**- PyTorch, TensorFlow, Scikit-learn, PEFT, RLHF, GRUs
- **NLP & LLM Frameworks**- LangChain, LangGraph, HuggingFace Transformers, Redis, AstraDB, Neo4j
- **Tools & Utilities**- Git, LaTeX, Monte Carlo simulation, Time-series analysis, GARCH models, Copula modeling